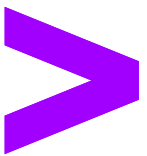


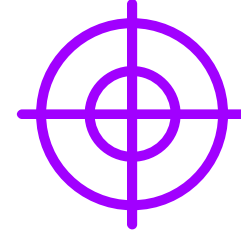
# Introduction to Selenium



# Module Objectives

At the end of this course, you will be able to:

- Understand about Selenium Suite and its tools
- Understand the features of Selenium
- Recognize Selenium as an automation library
- Understand about Selenium WebDriver & Grid
- Understand the purpose of Selenium IDE



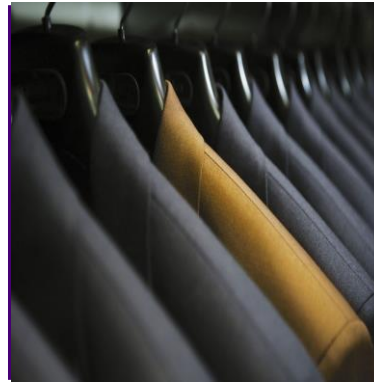
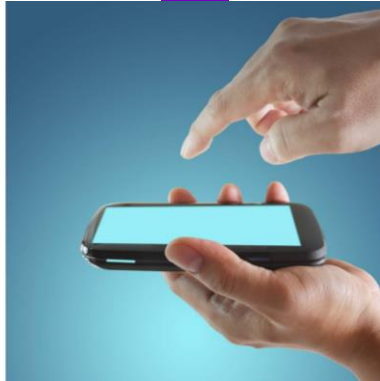
# Ground Rules

For a successful class, please:



Arrive on time

Keep your cell phones off



Wear business formals

Assist and show respect to your colleagues irrespective of skill and knowledge level



Do not use the web for personal reasons during training sessions

Adhere to attendance policy as directed by training coordinator



# Introduction to Selenium

- 01** Introduction to Selenium Suite
- 02** Selenium WebDriver
- 03** Selenium Grid
- 04** Selenium IDE

# Introduction to Selenium Suite

# Introduction to Selenium Suite (1)

## What is Selenium?

Selenium is to automate web browsers

Selenium is open source and available as a tool/library

Primarily Selenium is for automating web applications in functional testing, but certainly not limited to that

To get started with Selenium we have the below components



Selenium WebDriver



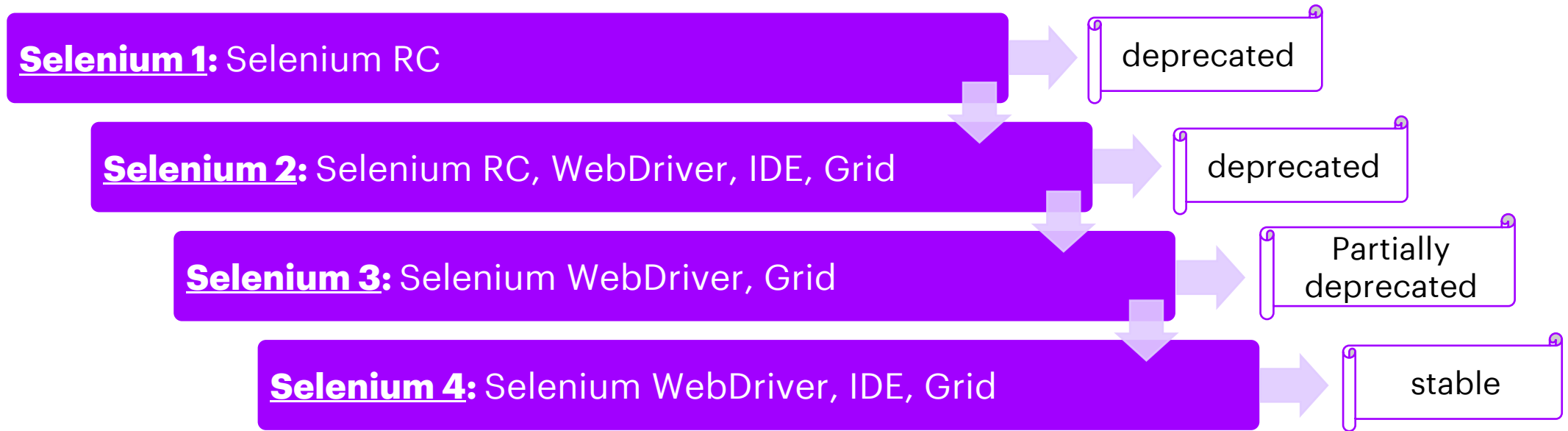
Selenium Grid



Selenium IDE

# Introduction to Selenium Suite (2)

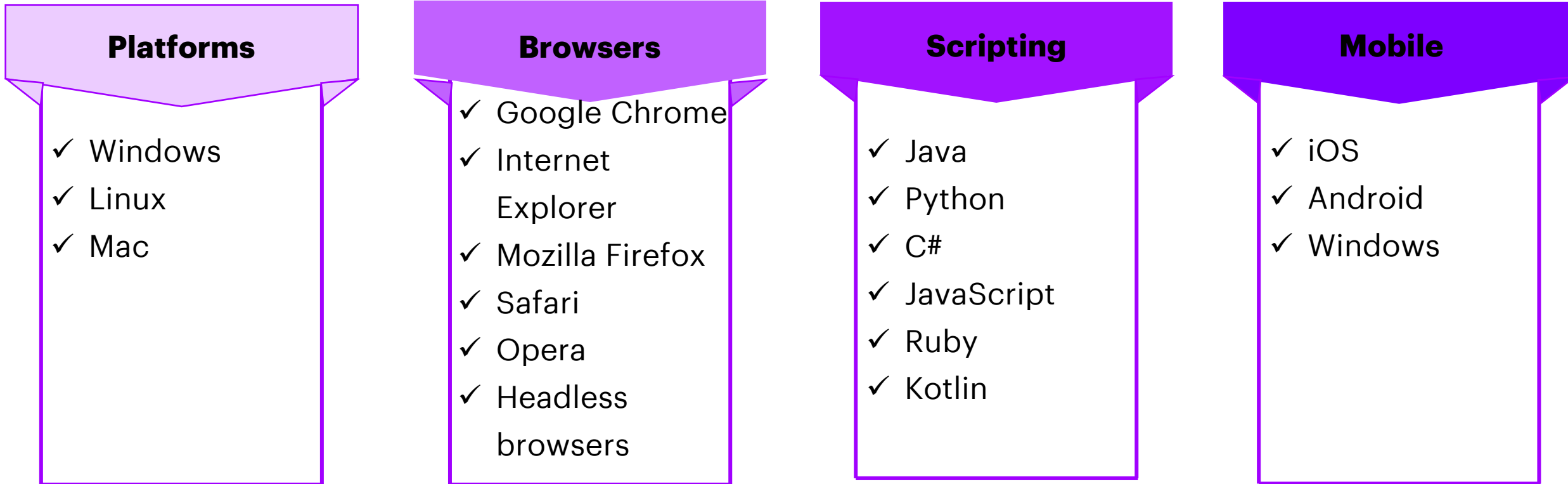
## Evolution of Selenium



- **Note:** Selenium Remote Control (RC) was the main Selenium project for a long time before the WebDriver/Selenium RC was merged to bring up Selenium 2. Selenium WebDriver was the successor of Selenium RC, thereby removing support for RC from Selenium 3. **This course focusses more on web automation using Selenium 4.**

# Introduction to Selenium Suite (3)

## Features of Selenium



**Note:** Selenium is supported by an active open community with the latest version of Selenium 4. All the latest information on Selenium can be accessed from <https://www.selenium.dev/>

# Selenium WebDriver

# Selenium WebDriver (1)

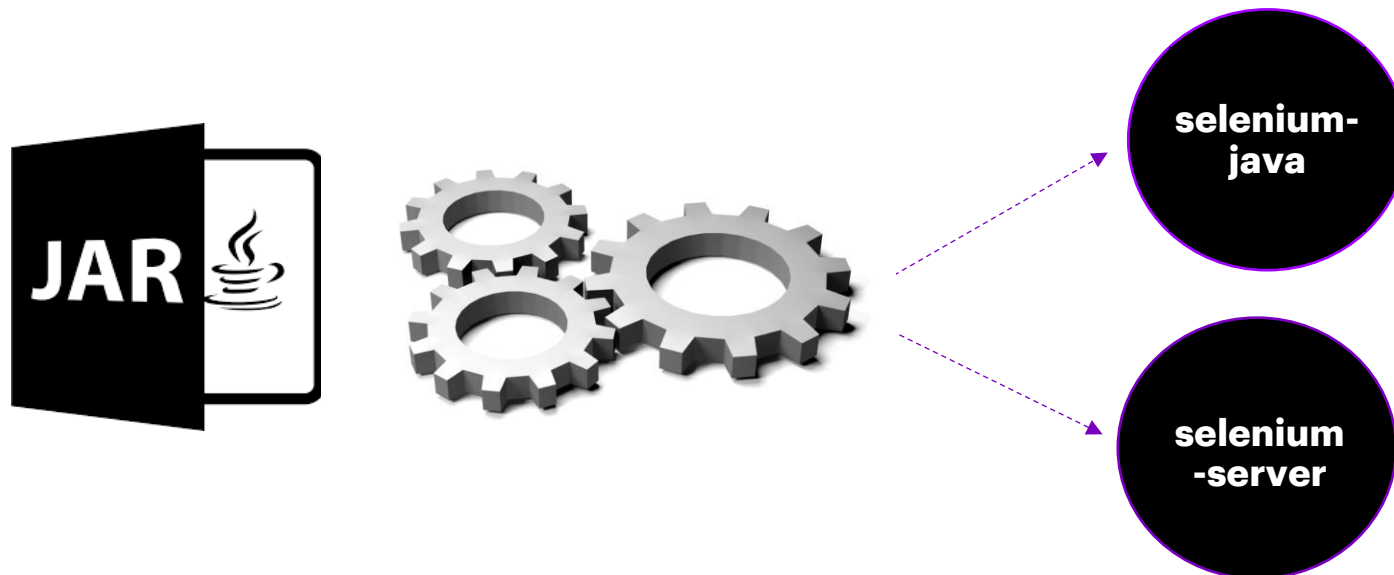
## Selenium Library (a.k.a. WebDriver)

Selenium WebDriver is a robust library with a collection of object-oriented APIs used for Document Object Model (DOM) interaction and browser control

The benefit of having Selenium as a library is that it supports scripting in popular OOP languages like java, c#, python

Selenium Library in Java is wrapped in the form of java archive files (.jar)

Selenium Java Library is available in 2 variants



# Selenium WebDriver (2)

Selenium Library (a.k.a. WebDriver)

## selenium-java as Library

- Using the selenium-java library, we can run test scripts only in the local machines
- It has only client-side bindings

## selenium-server as Library

- Using the selenium-server library, we can run test scripts in remote machines as well as in local machines
- It has built-in grid capabilities which follow a client-server architecture

# Selenium Grid

# Selenium Grid (1)

## Grid for Distributed Test Execution

Selenium Grid is a tool used to create an environment for distributed test execution

It is specialized in executing multiple WebDriver test scripts across:

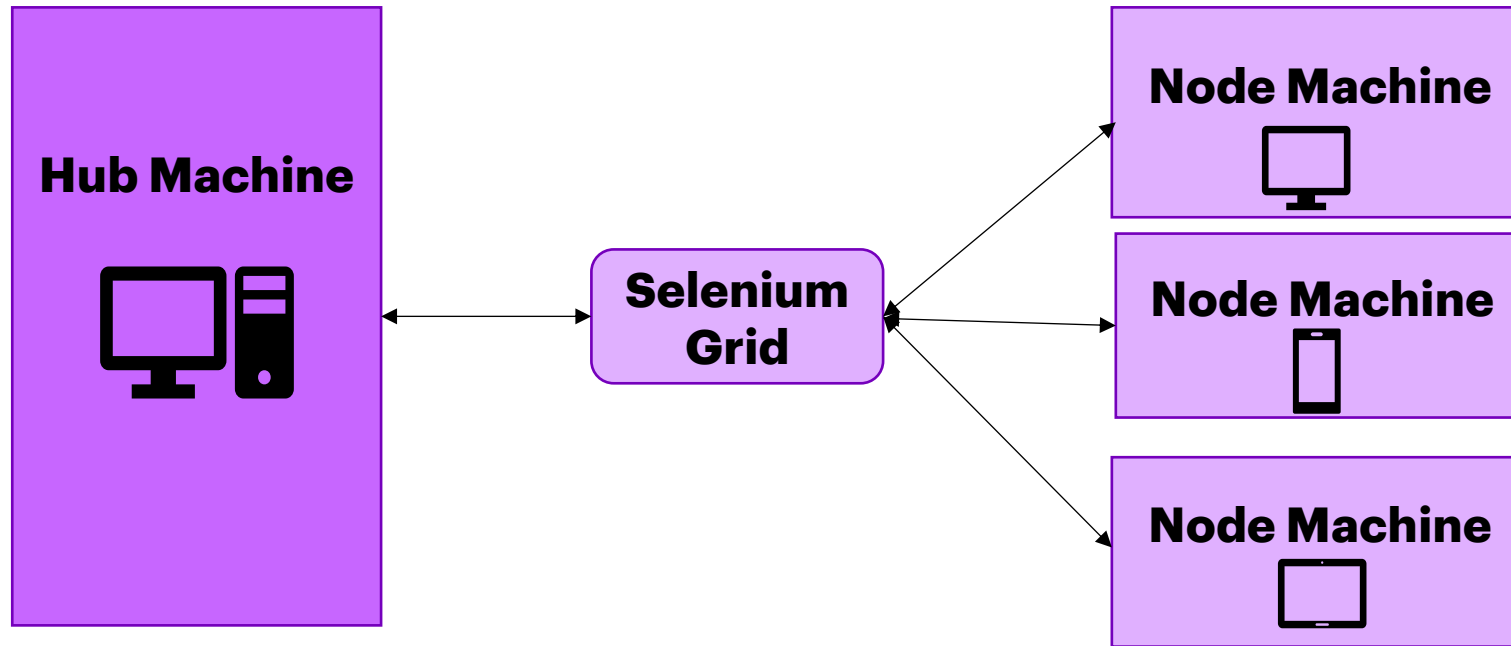
- Multiple browsers
- Multiple platforms
- Multiple remote machines

Benefits of using Selenium Grid:

- Parallel test execution across remote machines
- Reduced batch processing time
- Faster execution of large test suites
- Cross-platform testing
- Manage browser versions and configurations centrally

# Selenium Grid (2)

## Grid(hub-node) Architecture



### Note:

- **One hub** can be configured with **many nodes**
- **One node** can be configured with only **one hub**

# Selenium IDE

# Selenium IDE (1)

## Selenium IDE as a tool

Selenium IDE is a Google Chrome, Mozilla Firefox & Microsoft Edge add-on that will do simple record-and-playback of user interactions on browsers

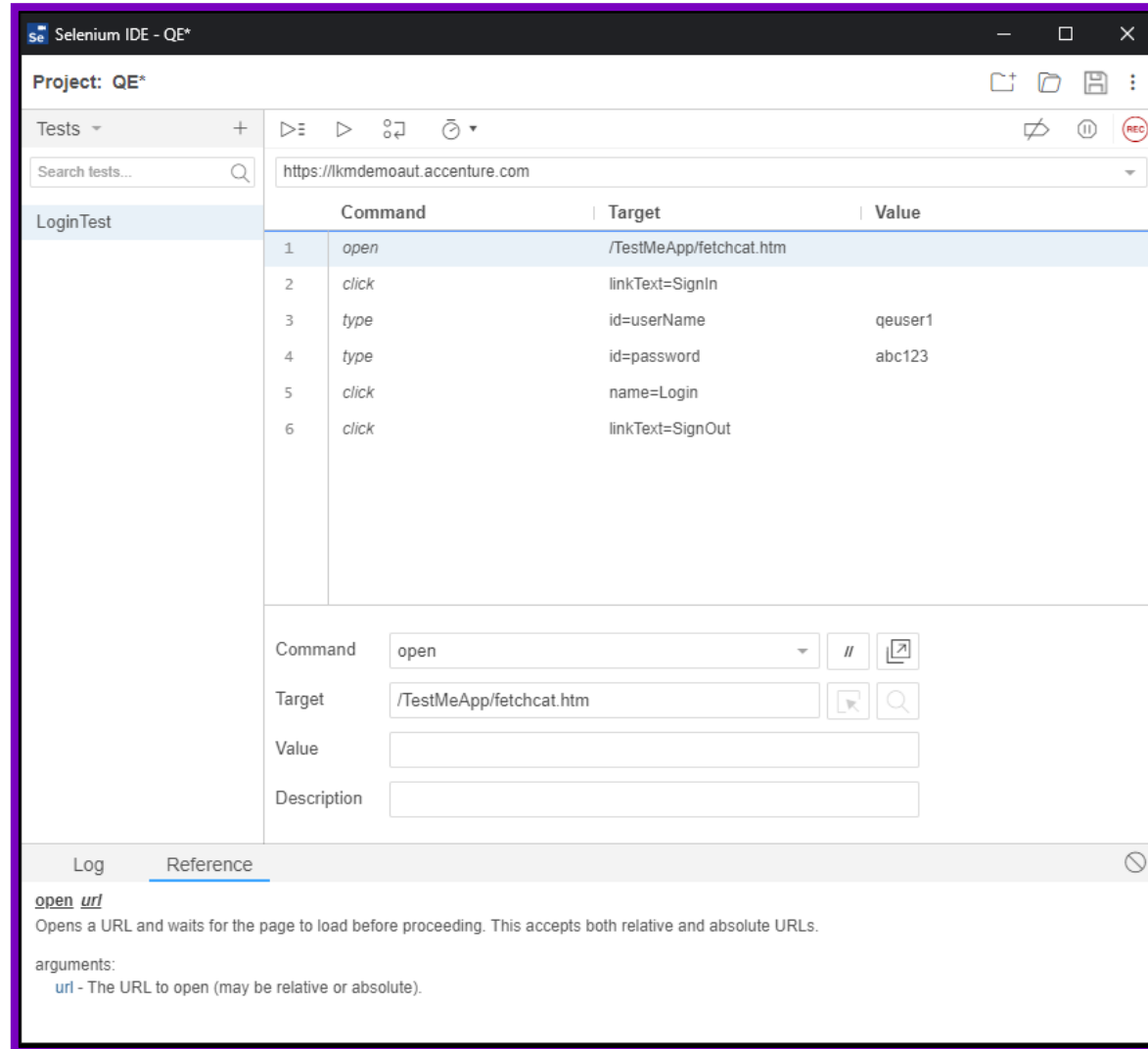
Selenium IDE is using the web-extension standards for modern browse like Google Chrome/Mozilla Firefox/Microsoft Edge

Selenium IDE is used to aid in automation based exploratory testing

Selenium IDE can create quick test scripts that can be used for simple web automation

# Selenium IDE (2)

## Selenium IDE look & feel



The screenshot displays the Selenium IDE interface for a project named 'QE\*'. The 'Tests' panel on the left shows a test named 'LoginTest'. The main area contains a table of commands for this test, with a URL bar at the top set to 'https://lkmdemoaut.accenture.com'.

|   | Command | Target                  | Value   |
|---|---------|-------------------------|---------|
| 1 | open    | /TestMeApp/fetchcat.htm |         |
| 2 | click   | linkText=SignIn         |         |
| 3 | type    | id=username             | qeuser1 |
| 4 | type    | id=password             | abc123  |
| 5 | click   | name=Login              |         |
| 6 | click   | linkText=SignOut        |         |

Below the table, the 'open' command is selected, showing its configuration fields: Command (open), Target (/TestMeApp/fetchcat.htm), Value (empty), and Description (empty).

At the bottom, the 'Reference' tab is active, showing the documentation for the 'open url' command:

**open url**  
Opens a URL and waits for the page to load before proceeding. This accepts both relative and absolute URLs.

arguments:  
url - The URL to open (may be relative or absolute).

# Knowledge Check

## Question 01



Which of the following components belong to the selenium suite?

a) Selenium WebDriver

b) Selenium IDE

c) Selenium Grid

d) All of the above

# Module Summary

Now, you will be able to:

- Understand about Selenium Suite and its tools
- Understand the features of Selenium
- Recognize Selenium as an automation library
- Understand about Selenium WebDriver & Grid
- Understand the purpose of Selenium IDE



**Thank You**

